

HDOC Day-38

Menu-Driven String Operations Using Separate Functions

Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

1. Count spaces, digits, and special characters in a string.
2. Count frequency of a given character in a string.
3. Toggle the case of each character in a string.
4. Exit.

Algorithm

1. Start the program and display a menu inside a loop.
2. Read the user's choice.
3. For choice 1, read a string and call a function that scans every character. Count spaces with a direct comparison, digits using `isdigit()`, and special characters as non-alphanumeric, non-space characters.
4. For choice 2, read a string and the required character. Call a function that compares each character with the required character and counts the matches.
5. For choice 3, read a string and call a function that changes lowercase letters to uppercase and uppercase letters to lowercase.
6. For choice 4, terminate the program.
7. For any other choice, display an invalid-choice message and show the menu again.
8. Stop.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void countCharacters(char str[])
{
    int spaces = 0, digits = 0, special = 0;
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
```

```

        if (str[i] == ' ')
            spaces++;
        else if (isdigit((unsigned char)str[i]))
            digits++;
        else if (!isalnum((unsigned char)str[i]))
            special++;
    }

    printf("Spaces = %d\n", spaces);
    printf("Digits = %d\n", digits);
    printf("Special characters = %d\n", special);
}

void countFrequency(char str[], char ch)
{
    int count = 0, i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] == ch)
            count++;
    }

    printf("Frequency of '%c' = %d\n", ch, count);
}

void toggleCase(char str[])
{
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (islower((unsigned char)str[i]))
            str[i] = (char)toupper((unsigned char)str[i]);
        else if (isupper((unsigned char)str[i]))
            str[i] = (char)tolower((unsigned char)str[i]);
    }

    printf("String after toggling case: %s\n", str);
}

int main(void)
{
    int choice;
    char str[200], ch;

    do
    {
        printf("\n--- MENU ---\n");
        printf("1. Count spaces, digits and special characters\n");
        printf("2. Count frequency of a character\n");
        printf("3. Toggle case of each character\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();
    }

```

```
switch (choice)
{
    case 1:
        printf("Enter a string: ");
        fgets(str, sizeof(str), stdin);
        str[strcspn(str, "\n")] = '\0';
        countCharacters(str);
        break;

    case 2:
        printf("Enter a string: ");
        fgets(str, sizeof(str), stdin);
        str[strcspn(str, "\n")] = '\0';
        printf("Enter character to find: ");
        scanf("%c", &ch);
        countFrequency(str, ch);
        break;

    case 3:
        printf("Enter a string: ");
        fgets(str, sizeof(str), stdin);
        str[strcspn(str, "\n")] = '\0';
        toggleCase(str);
        break;

    case 4:
        printf("Exiting program.\n");
        break;

    default:
        printf("Invalid choice. Please try again.\n");
}
} while (choice != 4);

return 0;
}
```

Sample Test Cases / Expected Output

Menu Choice	Sample Input	Expected Output
Operation 1	Input string: Hello 123!	Spaces = 1 Digits = 3 Special characters = 1
Operation 2	Input string: programming Character: r	Frequency of 'r' = 2
Operation 3	Input string: HeLLo 123!	String after toggling case: hElLo 123!
Operation 4	Choice: 4	Exiting program.

Result

The menu-driven C program was implemented successfully using separate functions for counting spaces/digits/special characters, finding the frequency of a character, toggling character case, and exiting the program.